

North pole.

- 25 C** Direction of magnetic flux density due to current in straight wire is determined by using RHGR. Plane of coils  $P$  and  $Q$  is parallel to the magnetic flux density due to current in straight wire, thus no magnetic flux link with coils  $P$  and  $Q$ . Plane of coil  $R$  is perpendicular to the magnetic flux density due to current in straight wire. Thus as the current in wire varies, there will be rate of change of magnetic flux linkage with coil  $R$ . Hence an e.m.f. is induced across the ends of coil  $R$ .
- 26 C** At  $t_1$  increasing current in coil  $O$  causes increase in  $B$  field generated by  $O$ .